

The MTUOC Project

Machine Translation at Universitat Oberta de Catalunya

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1. Overview of the MTUOC Project



Developed to facilitate the training and integration of **NMT** and **SMT** in professional environments.

- ▶ **The Challenge:** High technical skills required (Python/Bash), difficult integration with CAT tools, and heavy hardware requirements.
- ▶ **The Solution:** Well-documented scripts, graphical interfaces, and a configurable server.
- ▶ **Free/Open license:** Distributed under **GNU GPL license** to ensure technological independence.
- ▶ <https://mtuoc.github.io/>

2. Key Components



A comprehensive ecosystem covering the entire MT lifecycle:

Tools

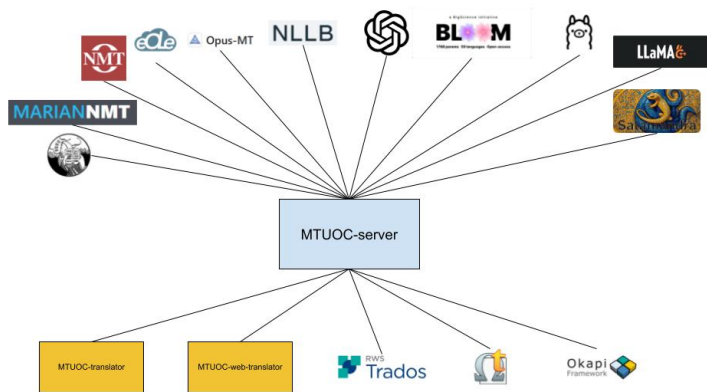
- ▶ **Corpus Creation:** Conversion (TMX, SDLTM, XLIFF) and alignment (Hunalign, SBERT).
- ▶ **Cleaning:** Filtering by length, language verification (fasttext), and rescoreing.
- ▶ **Training:** Scripts for Moses, Marian, OpenNMT, Eole, and fine-tuning for LLMs.
- ▶ **Clients:** Desktop and web applications for document translation.

3. The MTUOC-server



Acts as a bridge between the client (CAT tool) and the MT engine.

1. **Input:** Receives a segment from the CAT tool.
2. **Preprocess:** Tokenization, truecasing, SentencePiece.
3. **Translation:** Forwards to the MT engine or LLM.
4. **Postprocess:** Detokenization and undoing subwords.
5. **Output:** Returns the final text to the client.



4. Interoperability and Supported Systems



- ▶ **Protocol Mimicking:** Can behave like Moses, ModernMT, or OpenNMT servers to ensure CAT tool compatibility.
- ▶ **NMT Systems:** Supports Marian, OpenNMT, OpusMT, NLLB, CTranslate2, and commercial APIs (DeepL, Google).
- ▶ **LLM Integration:** Support for **Salamandra**, **Bloom**, **ChatGPT**, **DeepSeek**, and **Tower** via API or HuggingFace.
- ▶ **Custom Prompts:** Allows specific translation prompts and regex extraction.

5. Professional Workflows and Hardware



Advanced Features

- ▶ **Tag Restoration:** Algorithm to correctly place XML/HTML tags in neural translations.
- ▶ **TM Integration:** Returns matches from translation memories.
- ▶ **Plugins:** Ready for OmegaT and RWS Trados Studio.

Democratizing Access

- ▶ **Multiplatform:** Runs on Linux, Windows, and macOS.
- ▶ **Low Requirements:** No GPU needed for translation (only for training), allowing local and confidential work.